

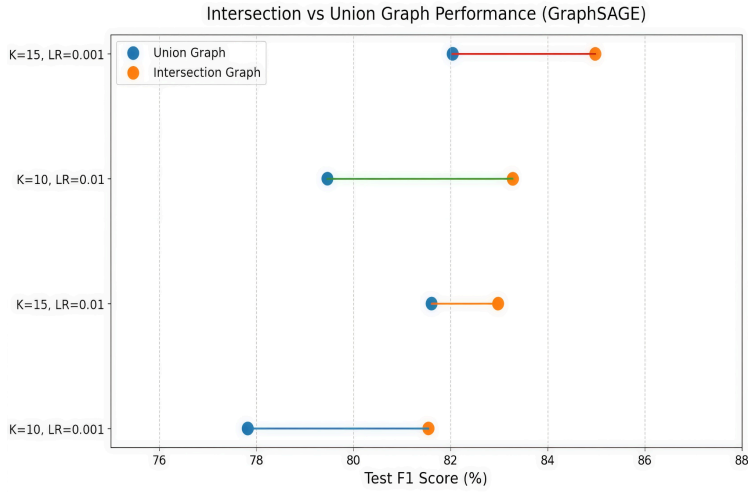
Supplementary Files (AttF-GNN)

1. Results After Varying the Encoder from GraphSAGE to GCN and GAT in the BRCA Dataset

Table 4: Performance comparison of GCN, GAT, and GraphSAGE under different hyperparameter settings

Top-K	LR	Model	Weighted F1
10	0.01	GCN	77.38%
10	0.01	GAT	79.71%
10	0.01	GraphSAGE	83.28%
10	0.001	GCN	79.55%
10	0.001	GAT	80.57%
10	0.001	GraphSAGE	83.36%
15	0.01	GCN	78.26%
15	0.01	GAT	81.37%
15	0.01	GraphSAGE	83.75%
15	0.001	GCN	81.65%
15	0.001	GAT	81.54%
15	0.001	GraphSAGE	85.60%

2. Evaluation of Union and Intersection Graph Strategies in Fusion Layer in the BRCA Dataset



3. F1-Score comparison between Euclidean distance and Cosine Similarity on the BRCA Dataset

Table 5: F1-Score comparison between Euclidean distance and Cosine Similarity.

Similarity Metric	C0	C1	C2	C3	C4
Euclidean Similarity	98.30	83.50	89.66	68.72	72.50
Cosine Similarity	97.30	83.50	87.57	68.72	72.50

4. Impact of Feature Selection

Supplementary Files (AttF-GNN)

Table 6: Feature Selection on TCGA-PRAD Dataset

Method	K	Accuracy	Weighted F1	Macro F1
ANOVA	128	0.944	0.92	0.61
ANOVA	264	0.916	0.91	0.57
ANOVA	512	0.963	0.96	0.79
Mutual Info	128	0.972	0.97	0.86
Mutual Info	264	0.963	0.96	0.82
Mutual Info	512	0.981	0.98	0.92

5. TCGA-GBM Result

The below table presents the comparison of AttF-GNN in the GBM cancer subtyping dataset. We accommodate only those architectures from the literature that were graph-based, having three omics into consideration and have publicly available code. For obtaining the result, we have considered 512 omics for both mRNA and DNA methylation. There were only 470 features available in the miRNA dataset. Finally, the highest weighted F1-score for AttF-GNN is 71.32% whereas for the same set up MOGLAM has 68.5% F1-score.

Table: Result on the TCGA-GBM Dataset

Method	Highest Achieved Weighted F1
SVM	61.77%
XGBoost	63.26%
Naive Bayes	54.53%
Decision Tree	64.72%
MOGONET	66.23%
MOGLAM	67.5%
AttF-GNN	71.32%

6. Considering More Feature Sets

For the BRCA Dataset, we have utilised more features, including 1000,1600 and 2000. Moreover, the performance does not increase, whereas the training time increases significantly.